Characterization and Modeling of 1.2 kV, 20 A SiC MOSFETs

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***Abstract* -- This paper presents a generic and complete process to characterize and model the newly developed silicon carbide (SiC) MOSFET. The static characteristics, including MOSFET I-V curves, body diode, nonlinear junction capacitances, as well as package stray inductances, have been fully characterized on a prototype 1.2 kV, 20 A SiC MOSFET under varying temperature from 25 °C to 200 °C. Characteristics particular to the SiC MOSFET and its advantages over the silicon counterparts are analyzed and explained. The switching performance of the device, on the other hand, has also been tested under room temperature using a specially designed double-pulse tester with minimized circuit parasitics. The characterization results are then used to build a SiC MOSFET model using the MOSFET modeling tool in Synopsys Saber. Finally, discussions are presented on how to improve the model accuracy in its switching behavior by obtaining static characteristics from switching waveforms.**

***Index Terms* -- Silicon carbide, power MOSFET, modeling**

I. INTRODUCTION

Silicon carbide (SiC) MOSFET has become one of the new wide bandgap devices which is promising to substitute silicon (Si) devices in the high-voltage, high-speed applications due to its higher breakdown voltage, lower on-state resistance, and better high-temperature operation capability [1-4]. Currently, laboratory prototypes of the new device are yet to be commercialized, which means manufacturer datasheets are still not available. Therefore, it is necessary to fully characterize the new SiC MOSFET in order to evaluate its potential utilization in converter systems and to compare it with its Si counterparts. The work conducted in this paper used a TO-247 packaged, 1.2 kV, 20 A SiC MOSFET developed by CREE [5]. The device's most important static characteristics, namely I-V curves, body diode, junction capacitances, etc, were all measured and their respective measurement methods described. The characterizations were also carried out under varying temperature from 25 °C to 200 °C, showing the superiority of the SiC device under much higher junction temperature than Si. The characterization process is generic so that it can be applied to the study of any new SiC MOSFET.

The switching characterization was also conducted under room temperature using a carefully designed double-pulse tester with minimized circuit parasitics to fully reveal the intrinsic switching performance of the device. Important quantities such as the switching times and losses were also recorded and are presented in this paper.

For the design and analysis of converters utilizing the new SiC MOSFET, a physics-based device model is also a necessity. The recently developed SiC MOSFET model presented in [6], however, is quite complicated in parameter extraction and subsequent coding in circuit simulators. The goal of this work, on the contrary, has been to use a mature Si MOSFET modeling tool to quickly build models for SiC MOSFETs. To this end, the Synopsys Saber Power MOSFET Tool has been used in this work successfully, and the paper presents its use as well as methods to improve the model accuracy in predicting SiC MOSFET switching waveforms.

II. STATIC CHARACTERIZATIONS

The static characterizations of SiC MOSFET include DC characteristics (I-V) measured with a Tektronix 371B curve tracer, and AC characteristics (impedances) measured using an Agilent 4294A impedance analyzer. The DC characteristics are represented by the leakage current I_{DSS}, output and transfer characteristics, on-state resistance $R_{DS(ON)}$, and body diode I-V curves. The AC characteristics include the internal gate resistance R_{GI}, nonlinear junction capacitances, as well as package parasitic inductances. These parameters are directly related to the performance evaluation and modeling of the device. For the temperature-dependent characterization, a hotplate was used to heat the MOSFET since neither of the above mentioned instruments can cope with elevated temperatures.

A. Leakage Current I_{DSS}

Same as the drain-source breakdown voltage BV_{DSS}, the leakage current is also an evaluation of the device blocking capability. However, I_{DSS} was adopted in this work instead of BV_{DSS} because it was measured under the rated voltage with the gate-source terminals shorted, hence avoiding the risk of destroying the device. Fig. 1 shows the measurement results, where a positive temperature coefficient for I_{DSS} can be inferred. As seen from the figure, the leakage current is well below 10 μA for the entire temperature range, which indicates a good blocking capability of the device at the rated voltage.

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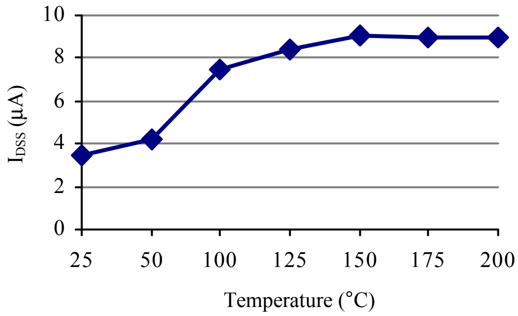


Fig. 1. Leakage current vs. temperature

B. Output and Transfer Characteristics

The output characteristics were measured under different gate voltage V_{GS} up to 15 V – the suggested drive voltage by the manufacturer. The transfer characteristic, on the other hand, was obtained under 15 V V_{DS} bias while varying V_{GS}. Fig. 2 plots these characteristics under 25, 100, 125 and 200 °C respectively. As seen from Fig. 2 (a), the trend of the output I-V curves of this 1.2 kV SiC MOSFET under varying temperature is quite different from what would be expected for Si MOSFETs. For $V_{GS} < 9$ V, the channel conductivity improves with the increasing temperature, while for $V_{GS} =15$ V the slope of the I-V curve first becomes steeper but then decreases again as the temperature increases. For the transfer characteristics shown in Fig. 2 (b), the threshold voltage $V_{GS(TH)}$ decreases with increasing temperature as expected, whereas the transconductance g_{fs} increases. This is again against the general knowledge about Si MOSFETs that g_{fs} should drop under higher temperature. These phenomena, however, can be explained from the variation trend of the on-state resistance about temperature which is to be stated in the next section.

C. On-state Resistance $R_{DS(ON)}$

The on-state resistance can be read directly from the output characteristic curves. In this work $R_{DS(ON)}$ was measured under V_{GS} = 15 V and I_D = 10 A. Fig. 3 shows the variation of $R_{DS(ON)}$ with regard to the temperature, from which a 5-8 times reduction in resistance can be observed compared to the state-of-art high-voltage Si MOSFETs with similar current rating.

Different from Si, the SiC MOSFET takes on an initial decrease in $R_{DS(ON)}$ with increasing temperature at low temperatures, then followed by an increase at higher temperatures. This is because of the different temperature dependence of three primary resistances within the device: channel resistance R_{CH}, JFET resistance R_{JFET}, and drift layer resistance R_{DRIFT}. R_{CH} decreases with increasing temperature due to the threshold voltage reduction and simultaneous increase in the channel mobility. R_{JFET} and R_{DRIFT}, on the other hand, increase as temperature rises due to the more active lattice vibrations. At low temperatures, R_{CH} decreases

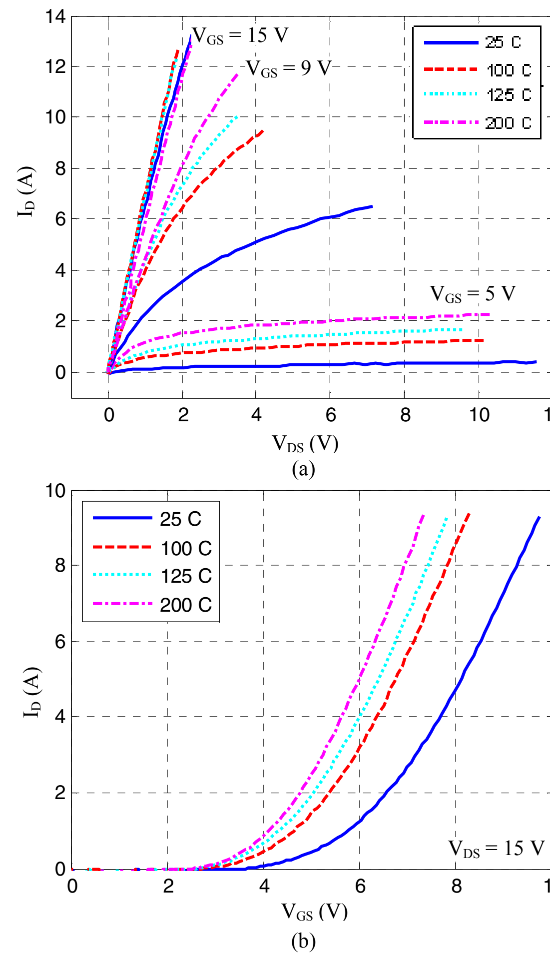


Fig. 2. Temperature-dependent output (a) and transfer (b) characteristics

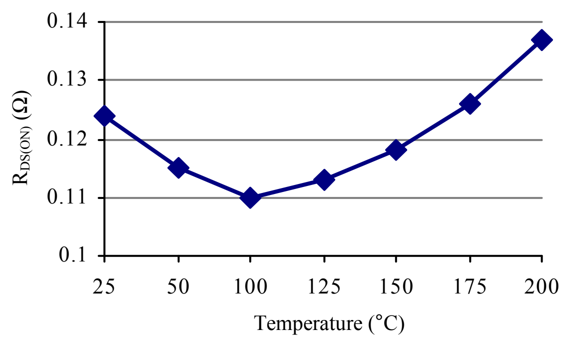


Fig. 3. On-state resistance vs. temperature

faster than the combined effect of R_{JFET} and R_{DRIFT}, causing the initial decrease in $R_{DS(ON)}$. At higher temperatures, the increasing effect of R_{JFET} and R_{DRIFT} dominates the overall resistance, resulting in the consequent increase in $R_{DS(ON)}$. The non-intuitive temperature-dependence of the output and transfer characteristics is thus explained.

It needs to be noted that the above phenomenon is only observed for 1.2 kV level devices, since their channel resistance is comparable to the JFET and drift layer resistances. For higher voltage rating SiC MOSFET (e.g. 10 kV), the much thicker drift layer makes R_{CH} negligible compared to R_{DRIFT} and thus gives a positive temperature coefficient for $R_{DS(ON)}$ at all temperatures.

D. Body Diode I-V Characteristic

The body diode I-V curves shown in Fig. 4 were obtained by reverse conducting the MOSFET with its gate and source terminals shorted. Less steep slants are observed as expected due to the increasing series resistance of the diode as the temperature rises. However, the forward turn-on voltage does not always decrease as expected, and an increase is observed for the 200 °C condition. This phenomenon is different from conventional Si MOSFETs and the mechanism behind it is still being explored.

E. Internal Gate Resistance R_{GI}

The internal gate resistance is a lumped resistance in series with the MOSFET gate terminal, representing the distributed resistor network connecting the individual cells within the device. Its value is important as it affects the time constant of the gate voltage and ultimately determines the switching speed limit of the MOSFET. However, this parameter is usually not provided in the datasheet. The measurement of R_{GI} is identical to the measurement of a capacitor equivalent series resistance (ESR), and was carried out with an impedance analyzer measuring the gate-source terminals while shorting the drain and source of the device. This setup measured the impedance of R_{GI} in series with the input capacitance C_{ISS}. Fig. 5 shows the measurement result, which depicts an increasing resistance of 0.4 to 0.55 Ω with the increasing temperature.

F. Nonlinear Junction Capacitances C_{ISS}, C_{RSS} and C_{OSS}

The junction capacitances, namely input capacitance C_{ISS}, reverse transfer capacitance C_{RSS}, and output capacitance C_{OSS}, are of vital importance to the MOSFET as they determine the dynamic behavior of the device during turn-on and turn-off transients. All these capacitances are strongly nonlinear functions of the drain-source voltage V_{DS}. For their measurement, the impedance analyzer alone is not sufficient as it cannot provide the voltage high enough to bias the device (only 40 V maximum). In this case, auxiliary circuits were adopted together with the impedance analyzer to extend the voltage bias to a much wider range. The circuits used in this work were modified from [7-8] and are illustrated in Fig. 6.

In the figure, "H" and "L" represent respectively the high potential/current terminal and low potential/current terminal of the impedance analyzer. The 1 μF capacitor connected to "H" isolates the instrument from the high voltage DC source. The measured capacitance is effectively the junction capacitance in series with this 1 μF isolation capacitance.

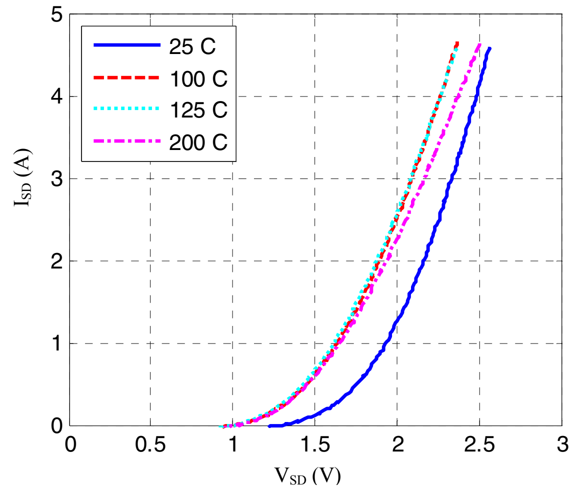


Fig. 4. Temperature-dependent body diode I-V curves

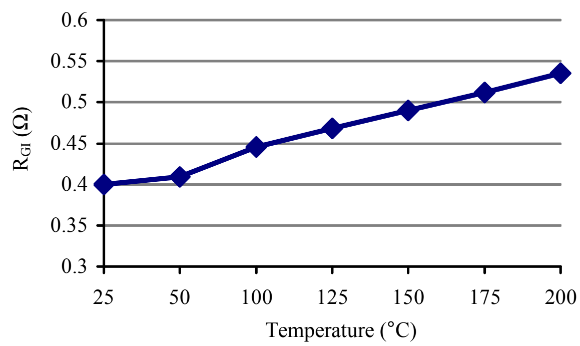


Fig. 5. Internal gate resistance vs. temperature

However, considering that the capacitances of interest are usually in the range of tens of pF to several nF, these measurements would still give very small error even without compensating off the series 1 μF capacitance.

The measurement results in Fig. 7 show that the junction capacitances drop drastically in the first 30 V and then decrease very slowly as the bias voltage continues rising. The bias voltage stopped increasing at 250 V since the trend of each capacitance-voltage (C-V) curve can already be seen clearly and extrapolation is quite easy. One can also observe from the figure that the C-V characteristics are insensitive to the changing temperature since all three sets of measurements overlap one another.

G. Package Parasitic Inductances

The device under study is in a TO-247 package. The parasitic impedances of the package can be lumped by three inductances L_G, L_D and L_S, which are in series with the gate, drain and source terminals respectively [9]. Basically three measurements are necessary to extract the stray inductances. Measuring the inductance between gate and source terminals with the drain floating gives the value of $L_G + L_S$. Likewise, measuring drain and source inductance gives $L_D + L_S$, and

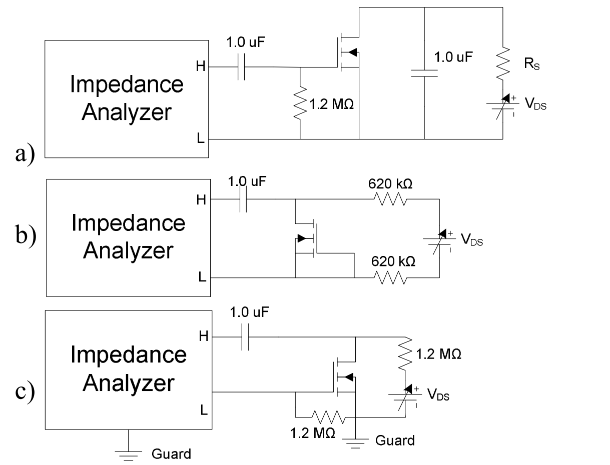


Fig. 6. Capacitance-voltage measurement setups. (a) C_{ISS}, (b) C_{OSS}, (c) C_{RSS}

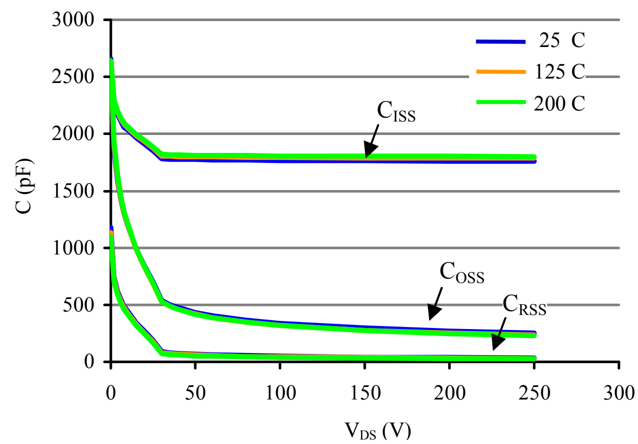


Fig. 7. Capacitance-voltage characteristics vs. temperature

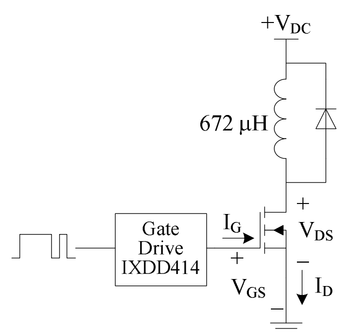


Fig. 8. Double-pulse tester schematic

measuring gate and drain gives $L_G + L_D$. The respective stray inductances can then be simply calculated from these measurements. In this case, the inductances were measured to be $L_G = 4.5$ nH, $L_D = 0.04$ nH, and $L_S = 3.2$ nH respectively.

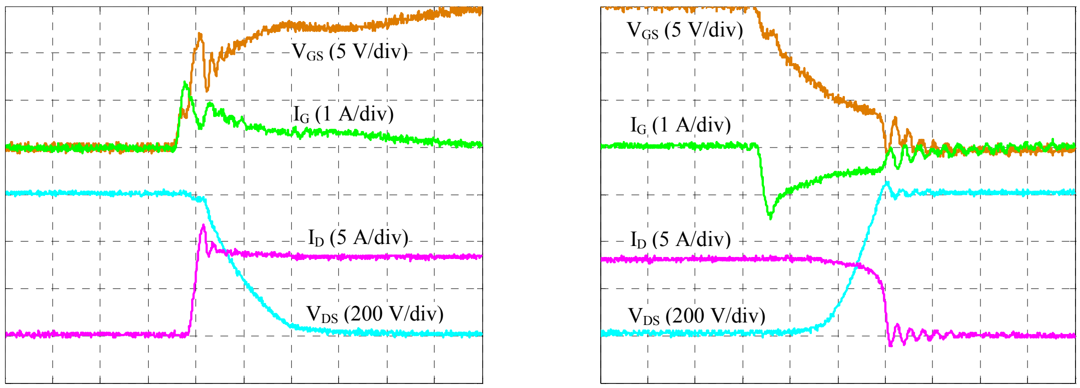
It also needs to be noted that the stray inductances are strongly affected by the lead lengths in the measurement. The above results were measured at the roots of the package leads. However, when measured at the tips of the leads, the inductances would be enlarged by almost six times. Therefore, when involving the package parasitics in the MOSFET model, it is important to relate the actual stray inductances to how the device is mounted on a printed circuit board.

III. Switching Characterization

The switching characteristics of the SiC MOSFET were also measured under room temperature with the use of a double-pulse tester whose schematic is shown in Fig. 8. The circuit layout of the tester was designed carefully to minimize the parasitics, so that the switching waveforms can reflect the device behavior as faithfully as possible. The gate drive chip used is IXYS IXDD414 which can achieve rise and fall times of 10 ns when driving 2 nF capacitive load. The drive voltage applied was 0-15 V. The freewheeling diode is a 1.2 kV, 20 A SiC Schottky barrier diode (SBD) prototype from Infineon. The use of the SBD eliminated the reverse recovery effect, and thus limited the current spike of the MOSFET during the turn-on process. The tests were done up to 600 V DC bus voltage, while varying the load current from 4 A to 10 A (considering margin for the current spike). The switching speed was adjusted by the use of different gate resistances of 10 Ω and 5 Ω respectively.

Fig. 9 shows the typical turn-on and turn-off transients of the gate voltage V_{GS} and current I_G, the drain-source voltage V_{DS}, as well as the drain current I_D. As seen from the figure, the rise and fall slew rates of I_D are quite fast compared to V_{DS}, achieving 6.5 ns during turn-on and 24.4 ns during turn-off, while it took 63.1 ns and 41.2 ns for V_{DS} to achieve full swing during turn-on and turn-off respectively. Also, compared with typical Si MOSFET gate voltage waveform, the SiC MOSFET does not have a very obvious "plateau" region in its V_{GS} waveforms. This is because the SiC MOSFET has a bigger channel-length modulation coefficient than the conventional Si device. Or, in another word, the I-V curves of the SiC MOSFET are not as flat as those of Si MOSFET in the saturation region, as can be inferred from Fig. 2 (a).

Switching times and losses were also recorded during the test. Fig. 10 exhibits the device total turn-on and turn-off times at 600 V as a function of the load current for different gate resistances. In this work, t_{on} is defined as the time from I_D reaching 10% of the steady-state current to V_{DS} falling to 10% of the DC bus voltage, while t_{off} is defined as from V_{DS} rising to 10% bus voltage to I_D falling to 10% load current. As seen, both the turn-on and turn-off times are approximately in linear proportion to the load current. The turn-on time increases with the load current, which is in accord with the common intuition, whereas the turn-off time decreases with the load current. This is because higher drain current leads to higher "plateau" voltage in V_{GS} during turn-off, increasing the gate discharge current and thus speeding up the turn-off process.



(a) (b)

Fig. 9. Typical turn-on (a) and turn-off (b) waveforms under 600 V, 8 A with $R_G = 10\ \Omega$. Time: 40 ns/div

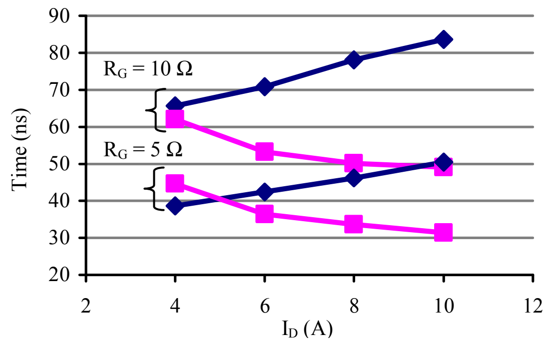


Fig. 10. Switching times vs. load current at 600 V.
Diamond markers: turn-on time t_{on}; square marker: turn-off times t_{off}

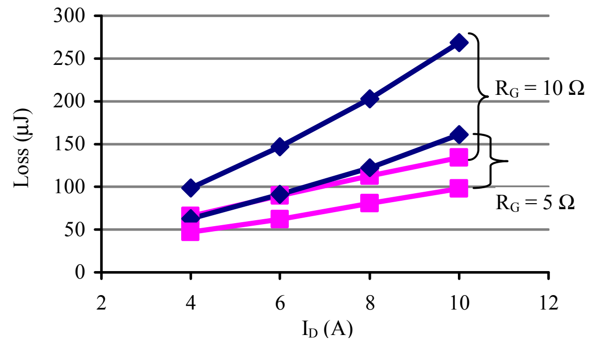


Fig. 11. Switching losses vs. load current at 600 V.
Diamond markers: turn-on loss E_{on}; square markers: turn-off loss E_{off}

Fig. 11 shows the switching losses versus drain current for different switching speed. Again the relationship between the switching energy and the load current is almost linear. Both the turn-on and turn-off losses increase with the load. However, E_{off} does not increase as fast as E_{on} because of the reduction in the turn-off time.

IV. SiC MOSFET Modeling

Although there has been a lot of effort in developing models for SiC MOSFET, and many of these models are claimed to match the switching waveforms quite well, yet the model parameter extraction process is still a tough job. Many SiC MOSFET models require physics-based parameters (e.g. doping density, electron mobility, etc), or special program dedicated to the parameter extraction, which are usually unavailable for the final device users [6][10-11]. Therefore, the goal of this work is to explore the possibility of building SiC MOSFET model based on the conventional static and switching characterization data, by the use of a mature Si MOSFET modeling tool. For this purpose, the Synopsys Saber Power MOSFET Tool [12] was used to build the model for this 1.2 kV SiC MOSFET.

The Saber Power MOSFET Tool generates the level-1 model which is well suited for examining the switching transients and losses of the power MOSFET. The tool requires the device information of output and transfer characteristics, internal gate resistance, nonlinear junction capacitances, body diode I-V curve and its junction capacitance, as well as package parasitic impedances, all of which can be obtained from the characterization process presented in Section II. All of these characteristic curves and parameters can then be imported into the modeling tool, and the model parameters can be adjusted visually to curve-fit the imported data [13].

Fig. 12 shows the curve fitting results of the MOSFET output and junction capacitance characteristics, which are apparently good predictions from the figure. However, in order to more accurately simulate the switching behavior of the device, measures should be taken beyond simply fitting the characteristic curves.

One problem lies in the curve-fitting of the output characteristics. Different from Si counterpart, the I-V curve of SiC MOSFET transits gradually from linear region to the saturation region [6]. Therefore, the device has not fully entered the saturation region even if the curve tracer has reached its maximum output voltage of 30 V. This trend can be inferred from Fig. 2 (a). Conducting the curve-fitting only in the linear and transition regions may lead to an erroneous saturation region characteristic of the model. In Fig. 12 (a) the mathematically optimal curve-fitting result gives the

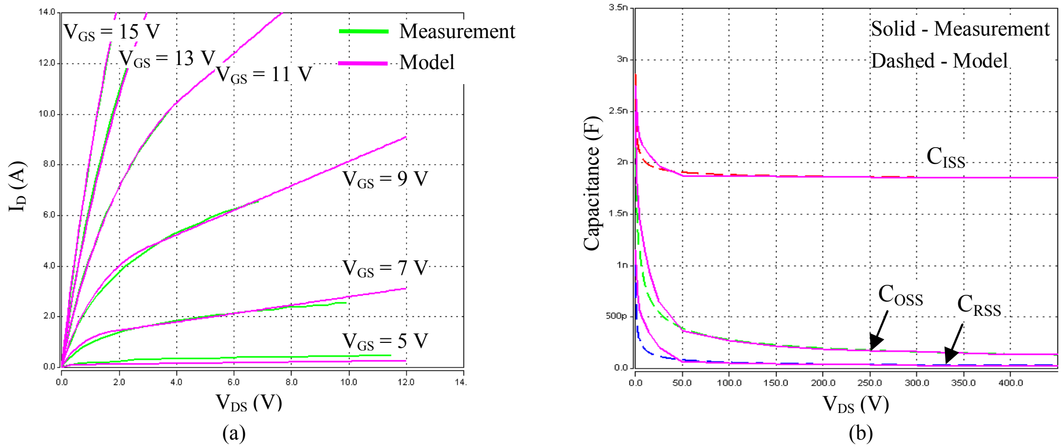


Fig. 12. Curve fitting results of (a) output characteristics, and (b) nonlinear junction capacitances

channel-length modulation parameter LAMBDA (i.e. the slope of the I-V curves in the saturation region) to be about 0.15, which is far too big for a reasonable LAMBDA value.

The parameter of LAMBDA is critical as it determines the shape of the saturation region. Simulations show that the model with better fitted saturation region can predict the switching waveforms more accurately. This is also intuitive because most part of the device switching trajectory would lie in the saturation region under the inductive load, hard switching condition. The major slew rates of both the voltage and current waveforms will be consequently determined by the shape of this region. Therefore, it is necessary to get a picture about how the saturation region looks like when it is not available from the curve tracer measurement.

The data needed for this is the transfer characteristic under high V_{DS} bias, which can be directly extracted by plotting I_D vs. V_{GS} from the switching waveforms. This follows from the MOSFET switching theory that when I_D rises and falls in an inductive load, hard switching condition, the device trajectory is well within the saturation region where the I_D vs. V_{GS} relationship represents the transfer characteristic under the corresponding V_{DS} bias [14]. To get this, the MOSFET should be switched very slowly by using a big gate resistance to minimize the circuit parasitic ringing and the currents flowing through the MOSFET junction capacitors. The I_D-V_{GS} curves obtained in this way, however, are different for turn-on and turn-off processes due to the voltage aberrance in V_{GS} waveforms caused by the common source inductance L_S, seen as the blue and yellow curves in Fig. 13. To compensate off the L_S effect, the real I_D-V_{GS} relationship can be derived as follows:

During turn-on:

$$v_{gs\_on} = v_{gi} + L_S \Delta I_d / \Delta t_r \tag{1}$$

During turn-off:

$$v_{gs\_off} = v_{gi} - L_S \Delta I_d / \Delta t_f \tag{2}$$

where v_{gs_on} and v_{gs_off} are the measured gate voltages, v_{gi} is the real gate voltage, Δt_r and Δt_f are the current rise and fall times, and ΔI_d is the switching current. It is assumed here that the drain current changes linearly during the switching transients, and that the gate current slew rate is much smaller than that of I_D so that the V_{GS} voltage aberrance is only determined by the drain current.

The real gate voltage is hence the weighted average of

$$v_{gi} = \frac{v_{gs\_on}\Delta t_r + v_{gs\_off}\Delta t_f}{\Delta t_r + \Delta t_f} \tag{3}$$

This method has been verified through Saber simulation. The "real" static transfer characteristic of the SiC MOSFET model under V_{DS} = 600 V was obtained from a DC sweep and plotted in Fig. 13 as the red dotted curve, while the green curve is the predicted one obtained from the simulation switching waveforms using (3). The close match of the two curves proves the effectiveness of the method.

The slow switching test was done using the same double-pulse tester shown in Fig. 8. The gate resistance was chosen to be 200 Ω to make the switching transients slow enough. Using the proposed method, the "predicted" transfer characteristics of the SiC MOSFET can be plotted in Fig. 14. As seen, the characteristic curves shift to the left as the bias voltage increases, indicating a bigger transconductance under higher V_{DS} voltage. This is reasonable for power MOSFETs in general. In this work, the additional transfer characteristic under 300 V was imported into the Power MOSFET Tool to adjust LAMBDA, which turned out to be 0.009 after compensation. Note that by doing this certain accuracy was sacrificed in the linear and transition region curve-fitting. However, as it will be shown later, better fitting in the saturation region of the model will still give a more accurate result in matching the experimental switching waveforms.

Another important parameter in MOSFET modeling is the Miller capacitance C_{RSS}. It was found from simulation that the MOSFET switching waveforms are extremely sensitive to the value of C_{RSS}, especially in the inductive load, hard switching

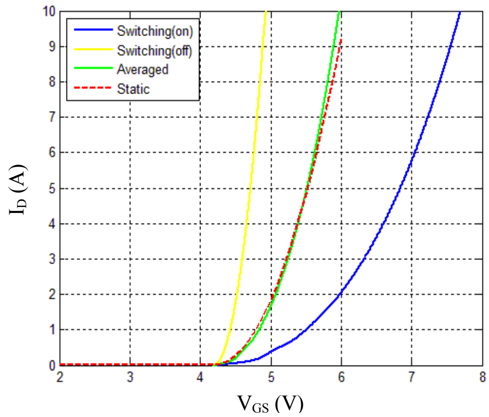


Fig. 13. Verification of transfer characteristic prediction in simulation

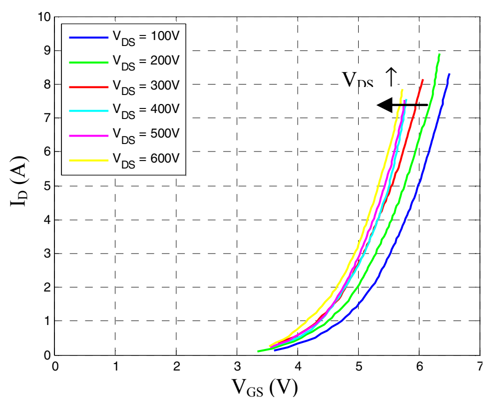


Fig. 14. Predicted transfer characteristics from slow switching test

condition, whereas the changes in C_{ISS} or C_{OSS} do not affect the switching too much. In principle, C_{RSS} is a capacitor connecting the input (gate) and the output (drain) of the MOSFET if one regards the device as a two-port network. According to the Miller Effect theory, this capacitor can be decoupled by putting two equivalent capacitors at the input and output ports separately, as shown in Fig. 15. From the input side, the decoupled capacitance is given by

$$C_{GDI} = C_{RSS}(1 - A_v) \tag{4}$$

where A_v is the AC amplification coefficient, given by

$$A_v = \frac{dV_{DS}}{dV_{GS}} = \frac{dV_{DS}}{dt} \Big/ \frac{dV_{GS}}{dt} \tag{5}$$

Because V_{GS} and V_{DS} always change in the opposite direction, therefore $A_v < 0$. During the switching process, when V_{DS} does not change, $A_v = 0$ and the input capacitance is simply C_{ISS}. However, during the V_{DS} rising or falling period, V_{GS} is effectively in its “plateau” stage while V_{DS} swings drastically between full bus voltage and nearly zero. This results in an extremely large A_v which magnifies C_{RSS} significantly and makes C_{GDI} dominate the input capacitance. Any small discrepancy in C_{RSS} will be exaggerated by A_v,

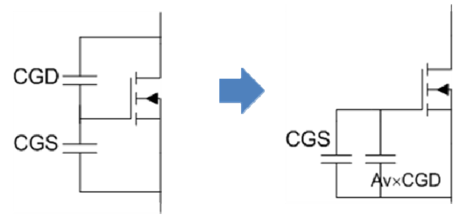


Fig. 15. Decoupling of the Miller capacitor.
(Only input capacitors are displayed)

leading to switching waveforms mismatch, especially V_{DS}. As a result, the seemingly perfect C_{RSS} curve-fitting does not guarantee an accurate prediction of switching waveforms, as even several pF changes in C_{RSS} can make a big difference in C_{GDI}. Experimental switching waveforms must be obtained so that the C_{RSS} C-V curve in the model can be adjusted finely to match the real switching condition.

Fig. 16 shows the comparisons of experimental and simulation waveforms under 600 V, 8A inductive load, hard switching condition with R_G = 10 Ω. This is also the case used to finely adjust C_{RSS}. Both simulation results with and without adopting transfer characteristic prediction are superimposed to show the difference. As seen from the figure, the MOSFET model considering the saturation region characteristics simulates the voltage and current slew rates much better than the one only fitting the linear region. The model was then used to predict another switching condition where the gate resistance was reduced to 5 Ω, as shown in Fig. 17. The simulation was not able to predict the high frequency ringing in the waveforms because the parasitic components in the circuit were not involved in the simulation. However the device model still simulates correctly the major characteristics of the switching waveforms, such as the V_{DS} and I_D slew rates, I_D and V_{GS} spikes during turn-on, as well as V_{GS} notch during turn-off.

The only discrepancies observed are the V_{GS} and V_{DS} waveform mismatches during turn-on process. As seen, the simulated V_{GS} seems to have a shorter “plateau” period thus leading to a faster V_{DS} falling speed. Looking back at the MOSFET modeling process one can find that only one transfer characteristic curve is allowed to be imported and fitted in the modeling tool, limited by the function of the program, which means the saturation region can only be partially fitted. Plus, the Si MOSFET model still has difficulty in fitting perfectly both the linear and saturation regions, and both the low current and high current regions at the same time in device I-V characteristics. Another possible reason for the V_{GS} mismatch is that the gate structure of the MOSFET is simplified as a single internal gate resistance in the model, whereas a more accurate circuit should be a RC ladder in fast switching condition, as multiple resonant points were observed beyond 50 MHz in the gate-source impedance measurement described in Section II (E).

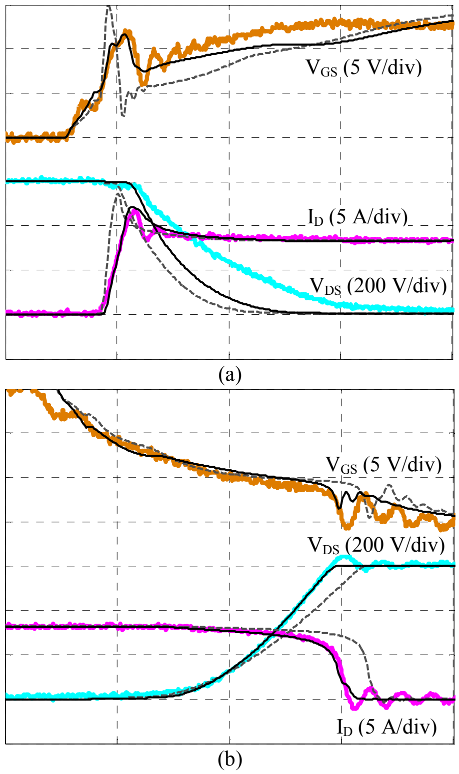


Fig. 16. Comparison of experimental (colored) and simulation (black and grey) waveforms with $R_G = 10\ \Omega$. (a) Turn-on, (b) turn-off. Time: 40 ns/div. Black – w/ transfer characteristic prediction; grey – w/o transfer characteristic prediction

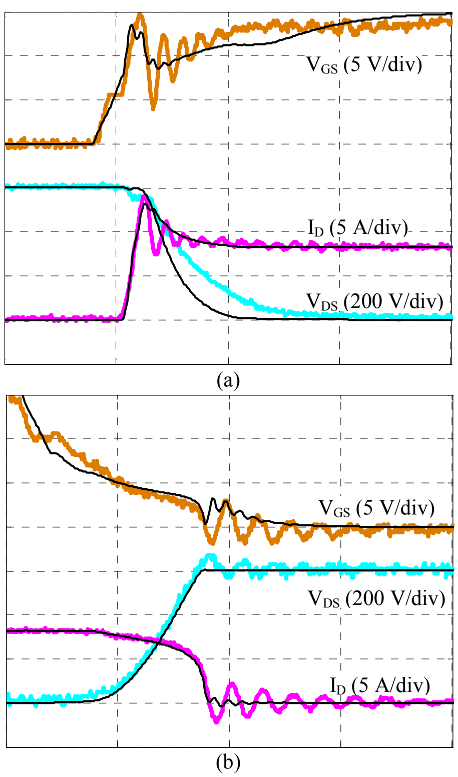


Fig. 17. Comparison of experimental (colored) and simulation (black) waveforms with $R_G = 5\ \Omega$. (a) Turn-on, (b) turn-off. Time: 40 ns/div

V. Conclusions

The full static and switching characterizations of a 1.2 kV, 20 A SiC MOSFET has been presented in this paper. The static characterization results showed the SiC device's superiority in blocking higher voltage while still keeping a very low $R_{DS(ON)}$ even under much higher junction temperature than Si MOSFET. The characterization process is generic so it can be applied to the study of any new MOSFET. The characterization data were also used to build a device model using the Si power MOSFET modeling tool. Discussions have been made on the most critical factors in the modeling – the saturation region characteristics and the Miller capacitance – in order to accurately predict switching waveforms from simulation. A method has also been proposed to obtain saturation region characteristics from switching waveforms. The developed model successfully simulated the real device switching behavior, although some discrepancies were also observed and explained. The discussions presented in this paper are very helpful for developing and building future models for SiC MOSFETs.

Acknowledgment

The work was supported by the Office of Naval Research under the Award No. N00014-07-1-0777.

References

[1] S. H. Ryu, A. K. Agarwal, J. Richmond, and J. W. Palmour, "4H-SiC DMOSFETs for high frequency power switching applications," in *Proc. Material Research Society Symp.*, vol. 764, pp. 69-74, 2003.

[2] S. H. Ryu, S. Krishnaswami, M. O'Loughlin, J. Richmond, A. Agarwal, J. Palmour, and A. R. Hefner, "10-kV 123-mΩ·cm², 4H-SiC power DMOSFETs," in *IEEE Electron Device Letters*, vol. 25, no. 8, pp. 556-558, Aug. 2004.

[3] S. H. Ryu, S. Krishnaswami, B. Hull, J. Richmond, A. Agarwal, and A. R. Hefner, "10 kV, 5 A 4H-SiC power DMOSFET," in *Proc. IEEE-ISPSD 2006*, pp. 1-4, 2006.

[4] T. McNutt, A. Hefner, A. Mantooth, and D. Berning, "Compact models for silicon carbide power devices," in *Solid-State Electronics*, vol. 48, pp. 1757-1762, 2004.

[5] http://www.cree.com/

[6] T. R. McNutt, A. R. Hefner, H. A. Mantooth, D. Berning, and S. H. Ryu, "Silicon carbide power MOSFET model and parameter extraction sequence," in *IEEE Trans. Power Electronics*, vol. 22, no. 2, pp. 353-362, Mar. 2007.

[7] International Rectifier, "Measuring HEXFET MOSFET characteristics," Application Note, AN-957.

[8] International Rectifier, "A more realistic characterization of power MOSFET output capacitance C_{OSS}," Application Note, AN-1001.

[9] E. McShane and K. Shenai, "RF de-embedding technique for extracting power MOSFET package parasitics," in *IEEE IWIPP 2000*, pp. 55-59, 2000

[10] M. Hasanuzzaman, S. K. Islam, L. M. Tolbert, "Design, modeling, and characterization of power MOSFET in 4H-SiC for extreme environment applications," in *Government Microcircuit Applications and Circuit Technology Conf. 2005*, pp. 449-452, 2005.

[11] P. D. Lungu, "Modeling of SiC vertical power DMOSFET structure using the fundamental MOSFET's set of equations," in *Int'l Semiconductor Conf. 1996*, vol. 2, pp. 327-330, 1996.

[12] "Synopsys Saber", Y-2006.06-SP2, Synopsys Inc., Mountain View, CA.

[13] Synopsys Saber, "Saber Model Architect Tool User Guide (Ver. Y-2006.06)", Chapter 8.

[14] S. Linder, *Power Semiconductors*, CRC Press, 2006.